

The Config Bench

Watt's Law — Power in DC Circuits | Rev 2.1

Episode TCB-PWR-001 | John King | @TheConfigBench | Target: ~12 min

Episode Overview (v2.1): Opens with giveaway winner announcement. Introduces Watt's Law and all derived forms. PERSONAL MOMENT in Segment 2: honest disclosure that the 47-ohm resistor was used in the early labs with a 9V battery — it didn't burn but got too hot to touch, and that was the trigger for creating this segment. NEW Segment 7: reverse formula $R = V^2 / P$ to find the correct safe resistor value. Closes with test questions and giveaway.

Episode Timing Overview

Segment	Content	Est. Time	Running
0	GIVEAWAY WINNER — previous video announcement	0:30	0:30
1	Welcome — bridge from Ohm's Law	0:30	1:00
2	What is Power? + personal story (hot resistor)	1:15	2:15
3	Watt's Law — all 4 formula forms (Xournal++)	1:30	3:45
4	Circuit 1 — single 47 ohm, power calculation	1:30	5:15
5	Circuit 2 — series resistors, power calculation	1:30	6:45
6	Circuit 3 — parallel resistors, power calculation	1:30	8:15
7	NEW: Choosing the correct resistor — $R = V^2 / P$	1:30	9:45
8	Real-world: wattage ratings, fuse sizing, LEDs	1:00	10:45
9	Test Questions x5 with answers	1:00	11:45
10	Outro — giveaway + subscribe CTA	0:30	12:15

SEGMENT 0 — GIVEAWAY WINNER ANNOUNCEMENT

SEGMENT 0 | Est. ~0:30

CAMERA: Medium shot at bench. Hold up winner card or display name as on-screen overlay.

Before we get into today's topic — we have a winner to announce from the last video.

If you left a comment on the Ohm's Law video, congratulations to [WINNER NAME] —

reach out to the email in the channel description and I'll get your prize sorted.

And if you missed it — stick around to the end of this one, because we're running it again.

ACTION: Fill in winner name before filming. Add as lower-third text overlay in post if preferred.

SEGMENT 1 — WELCOME & BRIDGE FROM OHM'S LAW

SEGMENT 1 | Est. ~0:30

CAMERA: Medium shot. Ohm's Law thumbnail visible on monitor or printed nearby.

Welcome back to The Config Bench — I'm John.

In the last video we covered Ohm's Law — V equals I times R .

Today we're adding one more tool to that kit: Watt's Law.

We're going to use it to figure out how much power our circuits are actually consuming — and how to know if a component is about to fail before it does.

SEGMENT 2 — WHAT IS POWER? + PERSONAL STORY

SEGMENT 2 | Est. ~1:15

PERSONAL NOTE: Honest personal disclosure — share this naturally, not as a confession. The tone should be 'here's what actually happened on this bench' not 'I made a mistake.' It builds trust and makes the lesson real.

CAMERA: Wide bench shot — power supply, multimeter, breadboard visible. Hold up the 47-ohm resistor.

Power is the rate at which energy is consumed. In a circuit, it's how much heat your components are generating.

And I want to be straight with you about something before we dive in.

When I was building the first labs on this channel — the Ohm's Law circuits — I was using a 47 ohm resistor with a 9 volt battery.

I ran through the math, got the current right, everything checked out on paper.

What I didn't do was stop and ask: how much power is that resistor actually dissipating?

It didn't burn up. But it got too hot to touch.

And that was the moment I realized I was only running half the math.

That's exactly what this segment is about — and honestly, that's why this segment exists.

Your power supply has a wattage rating. Your resistors have a wattage rating.

Your fuses are rated in amps — which is directly tied to watts.

If you skip Watt's Law, you won't always get a smoke show like in the Short. Sometimes you just get a component that's running way hotter than it should — shortening its life, introducing error into your measurements, or setting you up for a failure later.

That's what today is about — know your power, protect your components.

SEGMENT 3 — WATT'S LAW — ALL FORMULA FORMS

SEGMENT 3 | Est. ~1:30

WHITEBOARD (Xournal++): Write all four formula forms clearly. Circle $P = I \times V$ as the primary.

$$P = I \times V \mid P = V^2 / R \mid P = I^2 \times R \mid I = P / V$$

Here's Watt's Law: P equals I times V . Power equals current times voltage.

We can rearrange it — just like Ohm's Law — to solve for whatever we don't know.

$P = V$ squared divided by R is especially useful when you know voltage and resistance but haven't measured current yet.

We'll use all three power forms across today's circuits.

SEGMENT 4 — CIRCUIT 1: SINGLE 47-OHM RESISTOR

SEGMENT 4 | Est. ~1:30

QUCS: Open Circuit 1 — single 47 ohm, 9V. Show voltage and current readings.

Start simple — one 47 ohm resistor, 9 volt supply.

From Ohm's Law: current is 9 divided by 47 — about 191 milliamps.

Now Watt's Law: $P = I$ times V .

$$P = 0.191 \text{ A} \times 9 \text{ V} = 1.72 \text{ W}$$

1.72 watts. That resistor is rated at one quarter watt — 0.25 watts.

We are pushing nearly seven times its rated power through it.

Now you know why it got so hot in the lab. And why it burns in the Short.

WARNING: 1.72W into a 1/4W resistor = component stress or failure. Always verify wattage before powering up.

SEGMENT 5 — CIRCUIT 2: SERIES RESISTORS

SEGMENT 5 | Est. ~1:30

QUCS: Circuit 2 — 47 ohm + 100 ohm in series, 9V. Show total R, current, individual voltages.

Add a second resistor in series — 47 ohms and 100 ohms.

Total resistance: 147 ohms. Current: 9 divided by 147 — about 61 milliamps.

Power across the 47 ohm resistor: $P = I^2 \times R$.

$$P = (0.061)^2 \times 47 = 0.175 \text{ W}$$

0.175 watts — still under a quarter watt, but close.

Adding that series resistor dropped the current enough to keep us in the safe zone.

Calculate each component's power separately — they won't all be equal.

SEGMENT 6 — CIRCUIT 3: PARALLEL RESISTORS

SEGMENT 6 | Est. ~1:30

QUCS: Circuit 3 — 47 ohm in parallel with 100 ohm, 9V. Show branch currents.

Parallel is different — each branch sees the full 9 volts.

The 47 ohm branch draws 191 milliamps. The 100 ohm branch draws 90 milliamps.

$$P1 = 92 / 47 = 1.72 \text{ W} \mid P2 = 92 / 100 = 0.81 \text{ W}$$

Both resistors are exceeding their quarter watt ratings in this configuration.

Parallel circuits pull more total current — your power supply and your components both feel it.

This is why component ratings matter even more in parallel designs.

SEGMENT 7 — CHOOSING THE CORRECT RESISTOR VALUE [NEW]

KEY TEACHING MOMENT: Reverse the formula to find the MINIMUM safe resistance. This is the practical design skill that would have prevented the hot resistor in the early labs. Callback to the personal story in Segment 2.

CAMERA: Bench shot. Have resistors visible — 47 ohm, 100 ohm, 220 ohm, 330 ohm. Hold each up as referenced.

So — going back to what I just told you about those early labs.

The resistor didn't burn. But it got too hot to touch, and that told me something was wrong.

If I had run this calculation before I powered up, I would have caught it.

Here's how you find the RIGHT resistor value before anything gets hot.

We know our supply: 9 volts. We know our limit: one quarter watt.

We want the MINIMUM resistance that keeps us at or below that rating.

WHITEBOARD (Xournal++): Write the derivation step by step: $P = V^2/R$ rearranges to $R = V^2/P$

$$P = V^2 / R \text{ rearranges to } R = V^2 / P$$

Plug in our numbers: $R = 9 \text{ squared divided by } 0.25$.

$$R = 9^2 / 0.25 = 81 / 0.25 = 324 \text{ ohms (minimum safe value)}$$

324 ohms minimum. Our 47 ohm resistor is nowhere near that.

Resistors come in standard values — you won't find a 324 ohm in your kit.

So you go to the next standard value UP — 330 ohms.

$$\text{Verify: } P = 9^2 / 330 = 0.245 \text{ W — safely under } 1/4\text{W}$$

0.245 watts. Safe.

This is the calculation I should have run in Lab 1. Now you know it before you need it.

Always round UP to the next standard value. Never down.

WARNING: ALWAYS round UP to the next standard resistor value. Lower resistance = higher power = hot components or failure.

SEGMENT 8 — REAL-WORLD APPLICATIONS

SEGMENT 8 | Est. ~1:00

CAMERA: Show bench supply, fuse holder, resistor pack, LED as props.

This math applies everywhere on your bench.

Resistor wattage ratings — verify them before you power up. Every time.

Fuse sizing — amps times volts is watts. You're protecting a power budget.

LED current limiting resistors — LEDs are typically rated at 20 milliamps. $R = V^2 \text{ divided by } P$ sizes that resistor correctly every time.

Same formula, same process, every circuit.

SEGMENT 9 — TEST QUESTIONS

SEGMENT 9 | Est. ~1:00

Five questions — answers shown below. Pause if you want to work them out first.

Q1: A 9V source drives a 47-ohm resistor. What is the power dissipated?

- A) 0.19 W
- B) 0.86 W
- C) 1.72 W
- D) 3.44 W

Answer: C) 1.72 W — $P = 9^2 / 47 = 1.72 \text{ W}$

Q2: What is the minimum resistance to stay under 1/4W with a 9V supply?

- A) 47 ohms
- B) 147 ohms
- C) 270 ohms
- D) 324 ohms

Answer: D) 324 ohms — $R = 9^2 / 0.25 = 324 \text{ ohms}$

Q3: A circuit draws 0.5A from a 12V supply. What is total power consumption?

- A) 3 W
- B) 6 W
- C) 12 W
- D) 24 W

Answer: B) 6 W — $P = I \times V = 0.5 \times 12 = 6 \text{ W}$

Q4: You have a 330-ohm resistor on 9V. Is it safe for a 1/4W rating?

- A) Yes — $P = 0.245 \text{ W}$
- B) No — $P = 0.82 \text{ W}$
- C) Only in series
- D) Depends on temperature

Answer: A) Yes — $P = 9 \text{ squared} / 330 = 0.245 \text{ W}$ (under 0.25 W limit)

Q5: Two 100-ohm resistors in parallel on 12V — what is total power?

- A) 0.72 W
- B) 1.44 W
- C) 2.88 W
- D) 5.76 W

Answer: C) 2.88 W — Parallel $R = 50 \text{ ohms}$, $P = 12 \text{ squared} / 50 = 2.88 \text{ W}$

SEGMENT 10 — OUTRO & GIVEAWAY

SEGMENT 10 | Est. ~0:30

CAMERA: Medium shot. Smile. Giveaway item visible on bench.

That's Watt's Law — know your power, protect your components.

For the giveaway: leave a comment telling me what you'd use Watt's Law for on your own bench.

One winner picked from the comments — I'll announce it in the next video.

If this helped, hit subscribe — we're building from the fundamentals one video at a time.

I'm John. This is The Config Bench. See you next time.